Data Management (Packet Generator)
PC with PCIE
SFP+ TX
Ethernet Package
SFP+ RX
Ethernet Package
312.5 MHz
32 bit MAC + PCS/PMA
XGMII
Ethernet Frame
[31:0]
Packing and Segmentation
[31:0]
1900 byte
Ethernet to FSO Framer & Data Checker
Preamble 72 bits
Preamble [8:0]
ARQ TX Controller & Header Generator
Frame Header
[7:0]
14 byte
CRC-16
Frame Header
[7:0]
16 byte
IDLE
Multiplexer
Frame Header [7:0]
16 byte
312.5 MHz to 200 MHz
[31:0] to [63:0]
Clock Domain Crossing
(GTH to Logic clk)
[31:0]
1908 byte
CRC-32
FSO Payload
[31:0]
1912 byte
ARQ Retransmit RAM
FSO Payload
[31:0]
1912 byte
FIFO
FSO Payload
[31:0]
1912 byte
TXFN
IDLE Generator
PRBS [31:0]
1912 byte
FIFO
IDLE
Multiplexer
FSO Payload [31:0]
1912 byte
MGMT Payload
MGMT Generator
Telemetry [31:0]
1912 byte
FIFO
MGMT
Frame Type
FSO Header [7:0]
16 byte
RS(32,16)
[7:0]
32 byte
RS(255,239)
RS(255,239)
RS(255,239)
(Parallel 8 RS Codes)
FSO Payload [63:0]
1912 byte
[63:0]
2040 byte
Multiplexer
[63:0]
2072 byte
Scambler
[63:0]
2072 byte
LPC (24,16)
FSO Frame [95:0]
3108 byte
Multiplexer
[127:0]
3117 byte
[127:0] to [31:0]
200 MHz to 312.5 MHz
Clock Domain Crossing
(Logic clk to GTH)
Error Inject
[31:0]
312.5 MHz
64b/66b
Sync
Gearbox
GTH
10 Gbps
SFP+ Tx
FPGA TX - SDA FRAME ENCODER
10 Gbps
SFP+ Rx
GTH
[31:0]
3117 byte
[31:0] to [127:0]
312.5 MHz to 200 MHz
Clock Domain Crossing
(Logic clk to GTH)
[127:0]
Demultiplexer
Preamble 72 bits
FSO Frame [95:0]
3108 byte
LPC (24,16) Decoder
[63:0]
2072 byte
Descambler
[63:0]
2072 byte
Demultiplexer
[7:0]
32 byte
RS(32,16)
[63:0]
2040 byte
RS(255,239)
RS(255,239)
RS(255,239)
(Parallel 8 RS Codes)
FSO Header [7:0]
16 byte
FSO Payload [63:0]
1912 byte
[63:0] to [31:0]
200 MHz to 312.5 MHz
Clock Domain Crossing
(GTH to Logic clk)
[7:0]
16 byte
[31:0]
1912 byte
CRC-16 Checker
CRC-32 Checker
Header [7:0]
14 byte
Header Parser & ARQ RX Controller
CRC-32 Pass
ARQ Control
Timestamp
Satellite CPU
ACK / NAK
ARQ TX Controller
FSO Payload [31:0]
1908 byte
Frame Demultiplexer
FSO Payload [31:0]
1908 byte
ARQ RX Reorder Buffer
IDLE
Discard
MGMT
Telemetry / PNT
MGMT
Satellite CPU
FSO
Payload
[31:0]
1908 byte
Ethernet Frame Reassembly
Preamble [55:0] 7 byte
Start of Frame Delimiter (SFD) [7:0] 1 byte
Ethernet Preamble
[63:0]
8 byte
Ethernet
Payload
[31:0]
Multiplexer
Ethernet Frame
[31:0]
312.5 MHz
32 bit MAC + PCS/PMA
XGMII
SFP+ TX
Ethernet Package
SFP+ RX
Ethernet Package
PC with PCIE
FPGA RX - SDA FRAME DECODER